

FAULT CODES

Fault Code Chart - Control System

How to Retrieve Control System Fault Codes

Note: At least one fault code is present when the alarm at the platform controls produces two short beeps every 30 seconds for 10 minutes.

Note: Perform this procedure with the engine off, the key switch turned to platform controls and the red Emergency Stop button pulled out to the on position at both the ground and platform controls.

- 1 Open the platform control box lid.

WARNING Electrocutation/burn hazard. Contact with electrically charged circuits could result in death or serious injury. Remove all rings, watches and other jewelry.

- 2 Locate the red and yellow fault LEDs on the ALC-500 circuit board inside the platform control box. Do not touch the circuit board.

NOTICE Component damage hazard. Electrostatic discharge (ESD) can damage printed circuit board components. If the circuit board does need to be handled, maintain firm contact with a metal part of the machine that is grounded at all times when handling the printed circuit board OR use a grounded wrist strap.

- 3 **Determine the error source:** The red LED indicates the error source and will flash two separate codes. The first code will indicate the first digit of the two digit code, flashing once per second. It will then pause for 1.5 seconds and flash the second digit once per 0.5 second.

Note: When the red LED is flashing the code, the yellow LED will be on solid.

- 4 **Determine the error type:** The yellow LED indicates the error type and will flash two separate codes. The first code will indicate the first digit of the two digit code, flashing once per second. It will then pause for 1.5 seconds and flash the second digit once per 0.5 second.

Note: When the yellow LED is flashing the code, the red LED will be on solid.

- 5 Use the fault code table on the following pages to aid in troubleshooting the machine by pinpointing the area or component affected.

FAULT CODES

Error Source		Error Type		Condition	Solution
ID	Name	ID	Name		
21	Boom 1 Joystick (primary boom up/ down)	11	Value at 5V	Function is inoperative until joystick is calibrated. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
		12	Value too high		
		15	Value too low		
		16	Value at 0V		
		17	Not calibrated	Function is inoperative until joystick is calibrated.	Calibrate joystick
		18	Just calibrated	Initiate 1-second beep of alarm.	Self-clearing. (transient)
22	Boom 1 directional valves	21	Fault	Valve is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
23	Boom 1 flow control valve	12	Value too high	Valve is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
		15	Value too low		
		17	Not calibrated		
		18	Just calibrated	Normal function except threshold for one or both directions is zero.	Calibrate valve thresholds Self-clearing. (transient)
31	Boom 2 Joystick (secondary boom up/down or primary boom extend/retract)	11	Value at 5V	Joystick is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
		12	Value too high		
		15	Value too low		
		16	Value at 0V		
		17	Not calibrated	Function is inoperative until joystick is calibrated.	Calibrate joystick
		18	Just calibrated	Initiate 1 second beep of alarm.	Self-clearing. (transient)
32	Boom 2 directional valves	21	Fault	Valve is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.



Continued on next page

FAULT CODES

Error Source		Error Type		Condition	Solution
ID	Name	ID	Name		
33	Boom 2 flow control valve	12	Value too high	Valve is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
		15	Value too low		
		17	Not calibrated	Normal function except threshold for one or both directions is zero.	Calibrate valve thresholds
		18	Just calibrated		Self-clearing. (transient)
41	Turntable rotate joystick	11	Value at 5V	Joystick is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
		12	Value too high		
		15	Value too low		
		16	Value at 0V	Function is inoperative until joystick is calibrated.	Calibrate joystick
		17	Not calibrated		
		18	Just calibrated	Initiate 1-second beep of alarm.	Self-clearing. (transient)
42	Turntable rotate directional valves	21	Fault	Valve is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
43	Turntable rotate flow control valve	12	Value too high	Valve is operating outside of operational limits. Alarm sounds indicating fault.	Cycle power off, then on and problem should be corrected.
		15	Value too low		
		17	Not calibrated	Normal function except threshold for one or both directions is zero.	Calibrate valve thresholds.
		18	Just calibrated		Self-clearing. (transient)
44	Drive enable toggle switch	21	Fault	Drive enable function is inoperative.	Cycle power off, then on and problem should be corrected.

FAULT CODES

Error Source		Error Type		Condition	Solution
ID	Name	ID	Name		
51	Drive joystick	11	Value at 5V	Limited speed and direction frozen at zero and neutral. Alarm sounds indicating fault.	Power up controller with problem corrected.
		12	Value too high		
		15	Value too low		
		16	Value at 0V		
		17	Not calibrated	Joystick speed and direction frozen at zero and neutral.	Calibrate joystick
		18	Just calibrated	Initiate 1-second beep of alarm.	Self-clearing. (transient)
53	Drive flow valve (EDC)	12	Value too high	Limited speed and direction frozen at zero and neutral. Alarm sounds indicating fault.	Power up controller with problem corrected.
		15	Value too low		
		17	Not calibrated	Normal function except threshold for one or both directions is zero.	Calibrate valve thresholds
		18	Just calibrated		Self-clearing. (transient)
54	Drive brake valve	21	Fault	Limited speed and direction frozen at zero and neutral. Alarm sounds indicating fault.	Power up controller with problem corrected.
55	High speed drive motor Valve	21	Fault	Motor speed frozen in the low state. Alarm sounds indicating fault.	Power up controller with problem corrected.
61	Steer joystick	11	Value at 5V	Limited speed and direction frozen at zero and neutral. Alarm sounds indicating fault.	Power up controller with problem corrected.
		12	Value too high		
		15	Value too low		
		16	Value at 0V		
		17	Not calibrated	Joystick speed and direction frozen at zero and neutral.	Calibrate Joystick
		18	Just calibrated	Initiate 1-second beep of alarm.	Self-clearing. (transient)
62	Steer directional valves	21	Fault	Limited speed and direction frozen at zero and neutral. Alarm sounds indicating fault.	Power up controller with problem corrected.